

GUJARAT UNIVERSITY
K.S.SCHOOL OF BUSINESS MANAGEMENT
[FIVE YEARS' FULL-TIME M.Sc. (CA&IT) INTEGRATED DEGREE COURSE]

Project Title: Automatic Stair Light

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1. INTRODUCTION

1.1 OVERVIEW OF IOT

- The aim of the Automatic Stair Light system is to improve safety and save energy by automatically turning stair lights on when someone is using the stairs and turning them off when not needed.
- An Automatic Stair Light is a smart lighting system that turns on the lights automatically when it detects a person near or on the stairs. This makes it very useful at night or in dimly lit places, as it helps people see clearly and avoid accidents.
- The lights turn off by themselves when no one is around, which helps save electricity. This system is commonly used in homes, offices, and public buildings to make stairs safer, more convenient, and energy-efficient.

1.2 PROBLEM STATEMENT

- Many people forget to turn on stair lights at night or in dark places, which can cause accidents. Sometimes, they also forget to turn them off, which wastes electricity. Manually turning lights on and off is not always easy or convenient.
- So, we need a system that automatically turns the stair lights on when someone is near or using the stairs, and turns them off when not needed, to make stair use safer, easier, and more energy-efficient.

1.3 OBJECTIVE OF THE PROJECT

- The objective of the Automatic Stair Light system is to automate the lighting of staircases based on motion detection. The lights will turn on when movement is detected on the stairs and turn off after a specified time period without any movement.
- This system improves energy efficiency, enhances safety, and provides convenience in dark staircases, preventing accidents and conserving electricity.

1.4 SCOPE OF THE PROJECT

Safety Enhancement

- One of the primary scopes is to reduce the risk of accidents, especially in dark or dimly lit stairways.
- Helps elderly people, children, or guests safely navigate stairs at night without fumbling for a light switch.

Energy Efficiency

- Lights only turn on when motion is detected, thus reducing unnecessary electricity consumption.

- It supports automatic shut-off, ensuring that lights are not left on accidentally.

Smart Home Compatibility

- Can be easily integrated into smart home systems, supporting automation platforms like Google Home, Alexa, or other IoT ecosystems.
- Supports scheduling, remote access, and even app-based control.

Aesthetic and Design Flexibility

- Can be designed with LED strips or embedded lights to create a visually pleasing environment.
- Supports color-changing features and dimming options for ambiance control.

Low Maintenance

- The use of durable LEDs and reliable sensors means less maintenance over time.
- Most systems are designed for easy installation and long lifespan.

1.5 APPLICATION AND USE CASES

Residential Homes

- Used in homes to automatically light stairs during the night or when someone approaches.
- Especially helpful in homes with elderly people or children.

Apartment Complexes

- Installed in common stairways and corridors to improve safety and reduce energy consumption.
- Eliminates the need for continuous lighting.

Hotels and Hospitality

- Enhances guest experience by providing automatic lighting in stairwells and hallways.
- Adds a modern, premium touch to interior design.

Theaters and Cinemas

- Installed along stair edges to safely guide people without disrupting the movie with bright lights.

Hospitals and Healthcare Centers

- Helps patients and staff safely move during night hours or low-light conditions.

Shopping Malls and Commercial Buildings

- Used in emergency staircases and service areas for efficient and safe navigation.

Outdoor Public Areas

- Parks, railway stations, and flyovers use weatherproof versions for nighttime stair safety.

Smart Homes and Automation Projects

- Integrated into smart home systems as part of overall automation for convenience and energy saving.
-

Use Cases of Automatic Stair Light

Use Case	Description
Motion-activated stair lighting	Lights turn on only when someone is on or near the stairs.
Night-time safety	Prevents falls or injuries in dark staircases at night.
Power-saving feature	Reduces electricity usage compared to always-on lighting.
Hands-free operation	No need to touch switches – helpful when carrying items.
Child and elderly safety	Guides young and old users safely, especially in homes or care centers.
Aesthetic ambiance lighting	Creates a pleasing modern look using LED strips with dimming features.
Emergency backup lighting	Can be connected with backup batteries for use during power outages.

2. TOOLS AND COMPONENTS

2.1 LIST OF HARDWARE USED

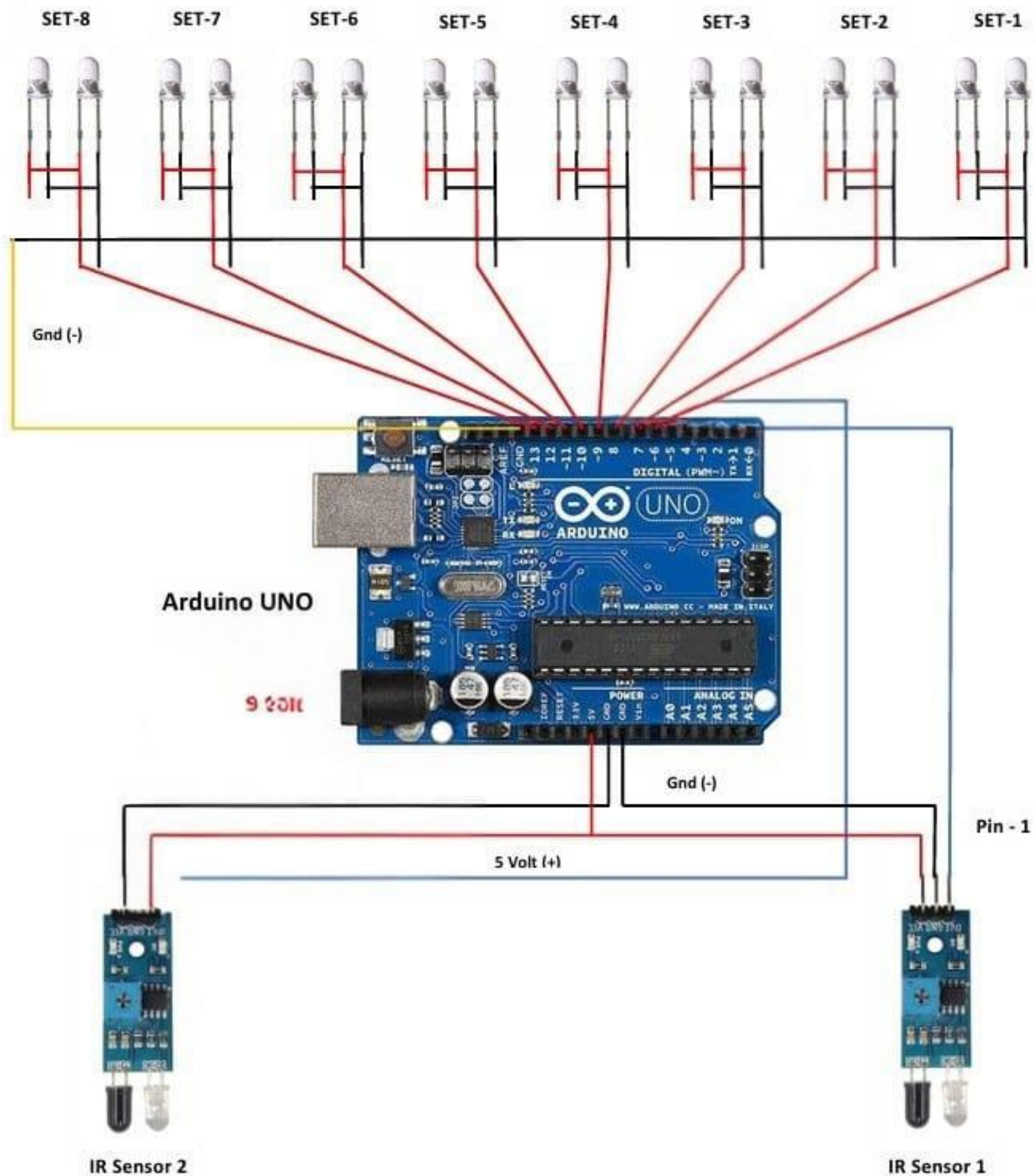
-  Arduino (Uno/Nano) or Microcontroller Board
 - Controls the logic of the system and responds to sensor inputs.
-  IR Sensors / PIR Motion Sensors
 - Detect motion or presence of a person near the stairs.
-  LED Strips or Individual LEDs
 - Provide the actual lighting along the stair steps.
-  Power Supply (e.g., 12V Adapter or Battery Pack)
 - Powers the LED strips and sensors.
-  Relay Module (Optional)
 - Allows control of higher-power AC lights instead of LEDs.
-  Wires and Connectors
 - For interconnecting all components.
-  Breadboard
 - For assembling the circuit temporarily or permanently.

2.2 LIST OF SOFTWARE USED

1. **Arduino IDE** – Used to write, compile, and upload code to the Arduino UNO.
2. **Serial Monitor (in Arduino IDE)** – For real-time monitoring and debugging.

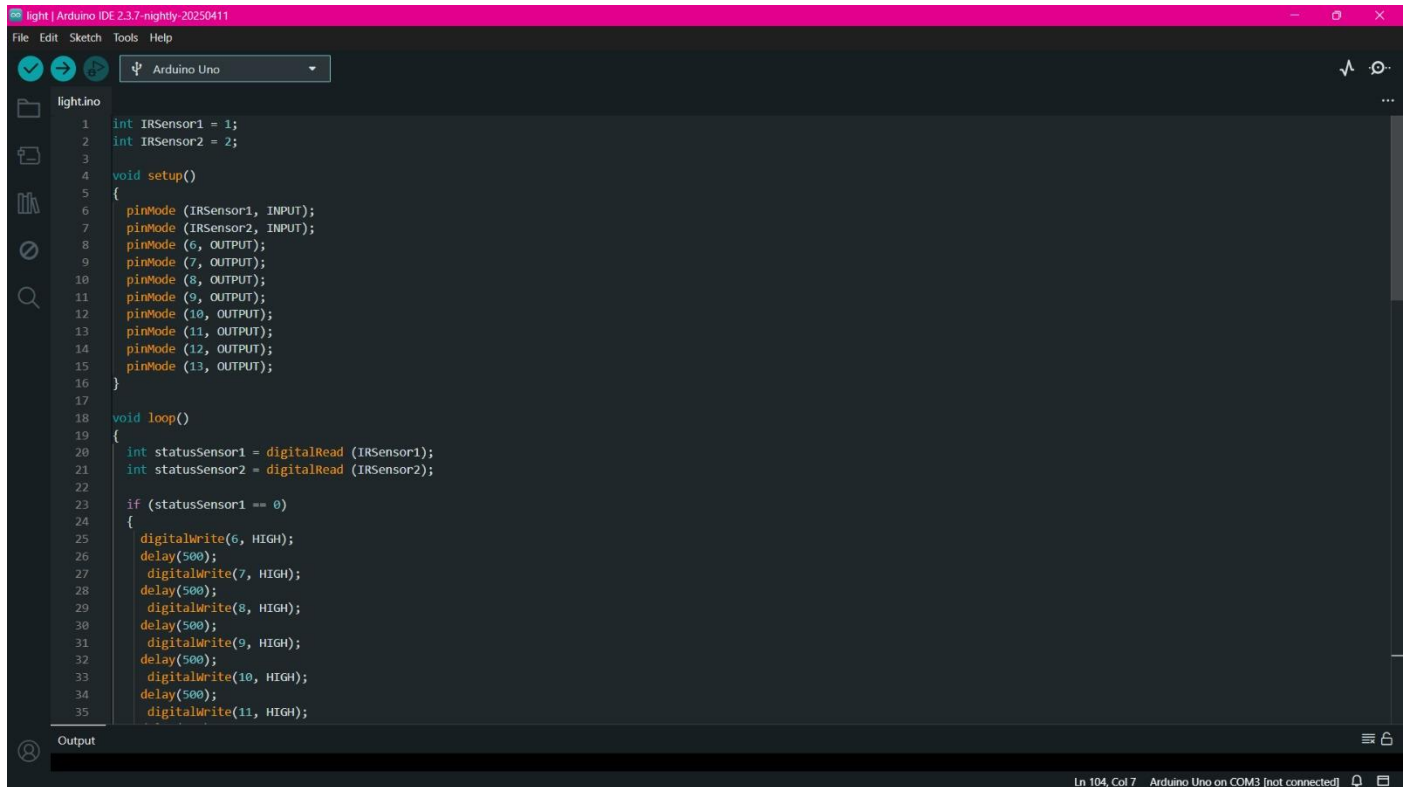
2.3 SIMPLE BLOCK DIAGRAM OR SETUP IMAGE

Staircase Light Circuit Diagram

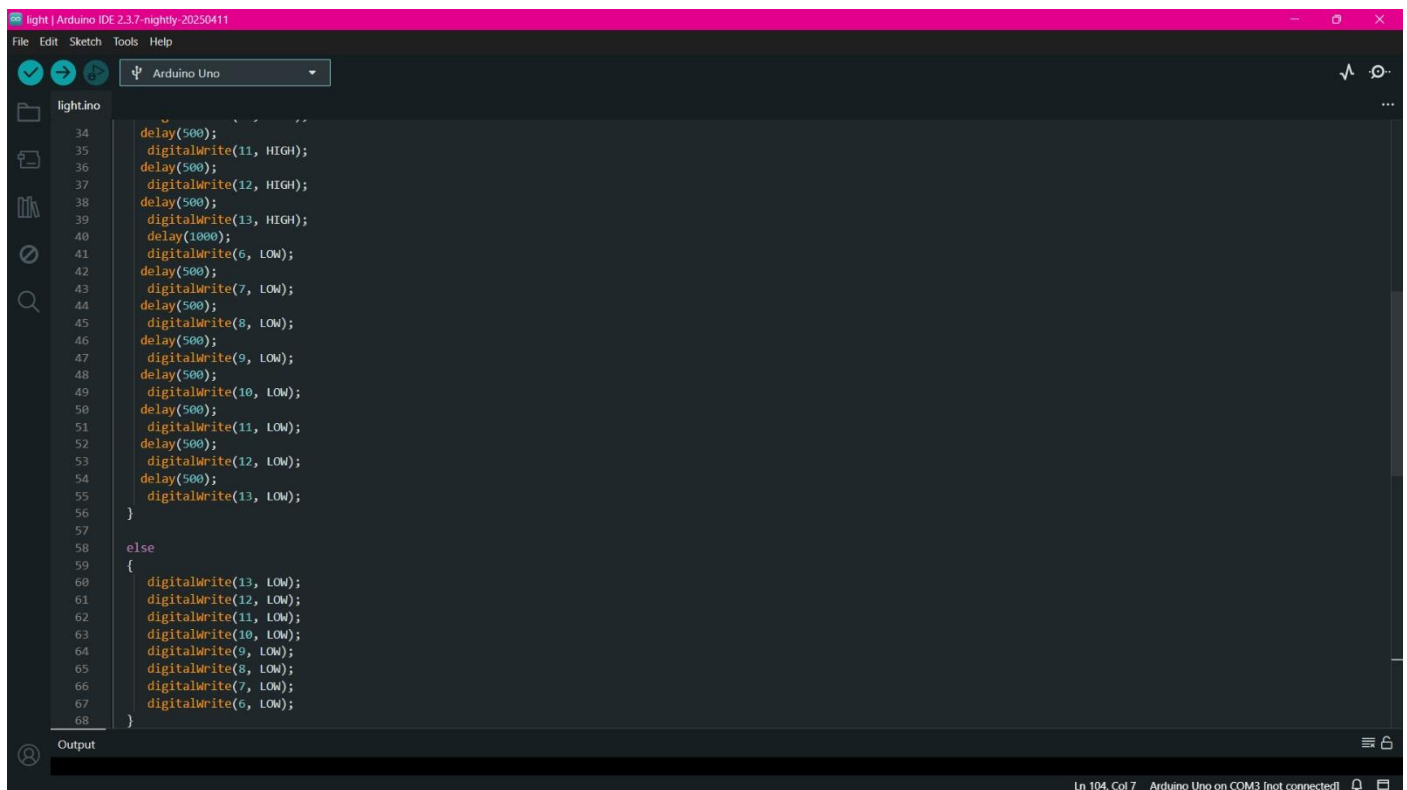


3. RESULT AND ANALYSIS

3.1 OUTPUT SCREENS



```
light.ino
1 int IRSensor1 = 1;
2 int IRSensor2 = 2;
3
4 void setup()
5 {
6   pinMode (IRSensor1, INPUT);
7   pinMode (IRSensor2, INPUT);
8   pinMode (6, OUTPUT);
9   pinMode (7, OUTPUT);
10  pinMode (8, OUTPUT);
11  pinMode (9, OUTPUT);
12  pinMode (10, OUTPUT);
13  pinMode (11, OUTPUT);
14  pinMode (12, OUTPUT);
15  pinMode (13, OUTPUT);
16 }
17
18 void loop()
19 {
20   int statusSensor1 = digitalRead (IRSensor1);
21   int statusSensor2 = digitalRead (IRSensor2);
22
23   if (statusSensor1 == 0)
24   {
25     digitalWrite(6, HIGH);
26     delay(500);
27     digitalWrite(7, HIGH);
28     delay(500);
29     digitalWrite(8, HIGH);
30     delay(500);
31     digitalWrite(9, HIGH);
32     delay(500);
33     digitalWrite(10, HIGH);
34     delay(500);
35     digitalWrite(11, HIGH);
```



```
34   delay(500);
35   digitalWrite(11, HIGH);
36   delay(500);
37   digitalWrite(12, HIGH);
38   delay(500);
39   digitalWrite(13, HIGH);
40   delay(1000);
41   digitalWrite(6, LOW);
42   delay(500);
43   digitalWrite(7, LOW);
44   delay(500);
45   digitalWrite(8, LOW);
46   delay(500);
47   digitalWrite(9, LOW);
48   delay(500);
49   digitalWrite(10, LOW);
50   delay(500);
51   digitalWrite(11, LOW);
52   delay(500);
53   digitalWrite(12, LOW);
54   delay(500);
55   digitalWrite(13, LOW);
56 }
57
58 else
59 {
60   digitalWrite(13, LOW);
61   digitalWrite(12, LOW);
62   digitalWrite(11, LOW);
63   digitalWrite(10, LOW);
64   digitalWrite(9, LOW);
65   digitalWrite(8, LOW);
66   digitalWrite(7, LOW);
67   digitalWrite(6, LOW);
68 }
```



```

light.ino
70  if (statusSensor2 == 0)
71  {
72      digitalWrite(13, HIGH);
73      delay(500);
74      digitalWrite(12, HIGH);
75      delay(500);
76      digitalWrite(11, HIGH);
77      delay(500);
78      digitalWrite(10, HIGH);
79      delay(500);
80      digitalWrite(9, HIGH);
81      delay(500);
82      digitalWrite(8, HIGH);
83      delay(500);
84      digitalWrite(7, HIGH);
85      delay(500);
86      digitalWrite(6, HIGH);
87      delay(1000);
88      digitalWrite(13, LOW);
89      delay(500);
90      digitalWrite(12, LOW);
91      delay(500);
92      digitalWrite(11, LOW);
93      delay(500);
94      digitalWrite(10, LOW);
95      delay(500);
96      digitalWrite(9, LOW);
97      delay(500);
98      digitalWrite(8, LOW);
99      delay(500);
100     digitalWrite(7, LOW);
101     delay(500);
102     digitalWrite(6, LOW);
103 }
104 else

```

Output

Ln 104, Col 7 Arduino Uno on COM3 [not connected]

```

light.ino
82      digitalWrite(8, HIGH);
83      delay(500);
84      digitalWrite(7, HIGH);
85      delay(500);
86      digitalWrite(6, HIGH);
87      delay(1000);
88      digitalWrite(13, LOW);
89      delay(500);
90      digitalWrite(12, LOW);
91      delay(500);
92      digitalWrite(11, LOW);
93      delay(500);
94      digitalWrite(10, LOW);
95      delay(500);
96      digitalWrite(9, LOW);
97      delay(500);
98      digitalWrite(8, LOW);
99      delay(500);
100     digitalWrite(7, LOW);
101     delay(500);
102     digitalWrite(6, LOW);
103 }
104 else
105 {
106     digitalWrite(13, LOW);
107     digitalWrite(12, LOW);
108     digitalWrite(11, LOW);
109     digitalWrite(10, LOW);
110     digitalWrite(9, LOW);
111     digitalWrite(8, LOW);
112     digitalWrite(7, LOW);
113     digitalWrite(6, LOW);
114 }
115 }
116

```

Output

Ln 104, Col 7 Arduino Uno on COM3 [not connected]

3.2 RESULT EVALUATION



3.3 ADVANTAGES OF PROPOSED SYSTEM

- Energy-efficient: Lights only activate when needed.
- Safety: Prevents falls on dark stairs.
- Convenience: No need to fumble for light switches.
- Cost Savings – Lower electricity bills and maintenance costs in the long run.
- Longevity of Lights – Reduces wear and tear by operating only when necessary.
- Modern look: Adds a stylish touch to the home.

4. FUTURE WORK

4.1 FUTURE ENHANCEMENTS

- Smartphone App Integration: For manual override and settings customization.
- Voice Assistant Control (Alexa/Google): Additional convenience.
- AI and Machine Learning: Predict usage patterns and optimize energy usage.
- Wireless Sensors: Reduce wiring complexity.
- Solar-powered models: For eco-friendly operation. Smarter sensors that can tell the difference between people and pets.